Michael J. Roosa

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EDUCATION

Doctorate of Philosophy: Physics **Conferred Dec. 12, 2024**
Thesis: Proton Removal Spectroscopy in TexAT.
Texas A&M University (TAMU), College Station, TX,

Masters of Science: Physics **Conferred May 11, 2023**
Texas A&M University (TAMU), College Station, TX,

Bachelor of Science: Physics **Conferred May 05, 2018**
Michigan State University (MSU), E. Lansing, MI,

PROFESSIONAL EXPERIENCE

Post Doctoral Researcher **Sept. 18, 2024 - Present**
Laboratoire Kastler Brossel (LKB)
Development and execution of new experiments deploying state-of-the-art quantum sensing x-ray microcalorimeters called Transition Edge Sensors (TESs) to measure antiprotonic atom x-ray cascades produced at CERN's Antimatter Decelerator (AD). This fundamental science effort will improve benchmarks for strong-field bound-state QED by two orders of magnitude while improving the technological readiness level of TESs through successive deployments in medium to high radiation environments.
Responsibilities have included:

- Operation of TES x-ray microcalorimeters and their supporting cryogenics: Adiabatic Demagnetization Refrigerators, Dilution Refrigerators.
- Supervising the research projects of Doctoral, Masters and Undergraduate students: UHV, Radiation Detection, Data Acquisition, X-ray and Gamma Spectroscopy.
- Acting as the zone safety officer for the TELMAX zone at the AD.
- Collaborating internationally as an affiliated visiting scientist at NIST (Boulder, CO).
- Presenting scientific results at conferences and workshops.
- Preparing technical reports, grant applications, and scientific publications.

Primary Investigators: Paul Indelicato & Nancy Paul

Graduate Research Assistant (*TAMU*) **Aug. 27, 2018 - July 03, 2024**
Designed, executed and analyzed radioactive ion beam experiments using the TEXas Active Target (TexAT) at the TAMU Cyclotron Institute. My work here focused on the development of novel charged-particle spectroscopy methods for nuclear structure studies. My work included large collaboration and individual experiments performed at multiple laboratories internationally.
Responsibilities included:

- Monte Carlo simulation of charged particle transport: *Geant4, Garfield++* for viability studies and as complement to experimental analysis.
- Operating and troubleshooting FPGA-based and traditional data acquisition systems.
- Performing terabyte scale data analysis: *Root (CERN)* packages.
- Developing and applying digital signal processing methods.
- Coordinating between external collaborators and local lab staff as well as facilitating their use of detector systems managed by my group.

Advisors: Gregory Christian & Grigory V. Rogachev.

Undergraduate Research Assistant (*MSU*) **Jan. 16, 2016 - Aug. 08, 2018**
Provided hands-on and simulation support (*Geant4, Garfield++*) for the design and commissioning the GAseous Detector with GErmanium Tagging (GADGET) detector system at FRIB.
Advisor: Christopher Wrede

AWARDS

Marie Skłodowska-Curie Post-Doctoral Fellowship *(LKB)* **Sept. 1 2026 - Present**

Awarded under the HORIZON TMA MSCA Postdoctoral Fellowships - European Fellowships action to support the Transition Edge Sensors for Antiprotonic Atom x-ray Spectroscopy (TESAAS) project (101281831).

EURO-LABS Transnational Access Grant *(LKB)* **2024 & 2025**

Granted to support test beams at affiliated research infrastructures. Supported by the European Union's Horizon Europe research and innovation program under grant agreement No 101057511.

Philip and Doris Moses Research Fellowship (*TAMU*) **2018 - 2022**

Merit-based fellowship for doctoral students in nuclear physics.

Hantel Endowed Fellowship (*MSU*) **2017 - 2018**

Merit-based fellowship for gifted undergraduates in physics.

TEACHING EXPERIENCE

Graduate Teaching Assistant *(TAMU)* **2019 & 2021**

Taught classroom and lab courses for university level physics classes.
Included training in physics education best practices and curriculum development.

Undergraduate Learning Assistant *(MSU)* **2015 - 2018**

Led recitations and tutoring sessions for university physics and mathematics classes.

MENTORING EXPERIENCE

Research Mentor *(LKB & TAMU)* **2022 - Present**

Helped to foster the development of graduate and undergraduate researchers with guidance and technical support.

Mentoring and Advising Graduates in an Inclusive Community **2022**

Founding Organizer and Mentor.

Conference Experience for Undergraduates APS DNP **2022**

Graduate Mentor to 4 undergraduate research attendees.

OUTREACH & MEDIA

CERN News: Antiprotons to test the Standard Model **June 2025**

News Article: The PAX experiment, the first user of the antimatter factory's new test beamline, is studying quantum electrodynamics using atoms containing an antiproton.

CERN News: An antimatter beam open for booking **June 2025**

News Article: Need an antiproton beam? TELMAX, the new test beamline at CERN's antimatter factory, is now open for booking.

Research Updates and Community Engagement **- Present**

Regular posts to my LinkedIn with research updates and insights as well as interaction with the broader physics community.

SERVICE

ASACUSA Collaboration - Deputy Group Leader **2026 - Present**

Assist in coordinating collaboration activities, communicating safety standards and best-practices and liaising between CERN and collaboration scientists.

Physics and Astronomy (P&A) Graduate Student Assembly **2019 - 2023**

Founded, lead and secured funding for this student advocacy and community organization.

Department of P&A Strategic Planning Retreat **2021**

Graduate Student Representative.

Dean's Graduate Student Advisory Council **2019 - 2023**

Represented P&A Graduate students to the College of Science.

Graduate and Professional Student Government **2018 - 2019**

Acted as elected senator representing the P&A Department.

SOFTWARE PROFICIENCIES

- *C++/C* and *Python* development in *UNIX* environments;
  specialized experience with *root* (CERN).
- Version control best practices with *Git*.
- *Geant4*: Monte Carlo particle tracking and physics simulation.
- High performance computing work load managers: *TORQUE*, *Apache SPARK*.

WORKSHOPS

2nd CERN Advanced Training School on Operation of Accelerators **2024**
Obtained hands-on training with operation of with CERN's accelerators: PS-Booster, ISOLDE, AD/ELENA and CLEAR Facilities.
(*CERN*)

DRD5 Quantum Sensing Autumn School **2024**
Investigated future research avenues made available by quantum sensing devices.
Included discussions and demonstrations concerned with emerging methods such as: Superconducting Sensors, Atom Interferometry and Chromatic Calorimetry.
(*CERN*)

Electron-Positron Pair Physics at the Cyclotron Institute **2024**
Explored and developed possible avenues of study considering electron positron pair generation at the TAMU Cyclotron Institute.
Included discussion of novel hardware developments and potentials theory contributions.
(*Texas A&M Cyclotron Institute*)

TPC2023 **2023**
Focused on developments in next generation TPC technology and addressing data acquisition need of the university and national lab nuclear physics TPC community.
(*Texas A&M Cyclotron Institute*)

National Nuclear Physics Summer School **2022**
Reviewed topics in theoretical and experimental nuclear physics, including:
hadronic spectroscopy and structure, QCD, heavy-ion physics, nuclear structure, nuclear astrophysics, fundamental symmetries, and neutrinos.
(*Massachusets Institute of Technology*)

CENTAUR Target-Making Pilot Workshop **2019**
Developed a pedagogy for preserving in-house target making methods for RIB experiments.
(*Louisiana State University*)

PUBLICATIONS

1. **M. Roosa**, G. Baptista, T. Azuma, D. Becker, F. Butin, O. Eizenberg, J. Fowler, H. Fujioka, D. Gamba, N. Garroum, M. Guerra, T. Hashimoto, T. Higuchi, P. Indelicato, J. Machado, K. Morgan, F. Nez, J. Nobles, B. Ohayon, S. Okada, S. Rathi, D. Schmidt, Q. Senetaire, J. Sommerfeldt, D. Swetz, J. Ullom, P. Yzombard, M. Zito, N. Paul, "First Deployment of a Quantum Sensing TES for Antimatter Studies". (*in preparation*)
2. K. Morgan, T. Azuma, G. Baptista, D. Becker, D. Bennett, F. Butin, J. W. Dean, O. Eizenberg, J. W. Fowler, H. Fujioka, D. Gamba, J. D. Gard, N. Garroum, M. Guerra, T. Hashimoto, T. Higuchi, M. Hori, P. Indelicato, M. Keller, J. Machado, J. A. B. Mates, N. Nakamura, J. Nobles, F. Nez, B. Ohayon, S. Okada, N. Ortiz, S. Rathi, **M. Roosa**, J. -Y. Rouss´e, T. Saito, D. Schmidt, Q. Senetaire, J. Sommerfeldt, P. Szypryt, J. Ullom, J. Weber, P. Yzombard, M. Zito, N. Paul, D. Swetz. A Transition-edge Sensor Spectrometer for Precision Spectroscopy of Antiprotonic Atoms
3. G. Baptista, S. Rathi, **M. Roosa**, Q. Senetaire, J. Sommerfeldt, T. Azuma, D. Becker, F. Butin, O. Eizenberg, J. Fowler, H. Fujioka, D. Gamba, N. Garroum, M. Guerra, T. Hashimoto, T. Higuchi, P. Indelicato, J. Machado, K. Morgan, F. Nez, J. Nobles, B. Ohayon, S. Okada, D. Schmidt, D. Swetz, J. Ullom, P. Yzombard, M. Zito, N. Paul, "Towards Precision Spectroscopy of Antiprotonic Atoms for Probing Strong-field QED", PoS(EXA-LEAP2024), 480, 085, July 2025.
4. E. Harris, M. Barbui, J. Bishop, G. Chubarian, Sebastian Konig, E. Koshchiy, K.D. Launey, Dean Lee, Zifeng Luo, Yuan-Zhuo Ma, Ulf-G. Meissner, C.E. Parker, Zhengxue Ren, **M. Roosa**, A. Saastamoinen, G. H. Sargsyan, D.P. Scriven, Shihang Shen, A. Volya, Hang Yu, G.V. Rogachev. "Quantifying alpha clustering in the ground states of ^{16}O and ^{20}Ne", arXiv:2507.17059v1 (*in submission*)

5. S. Ota, P. Capel, G. Christian, V. Durant, K. Hagel, E. Harris, R. C. Johnson, Z. Luo, F. M. Nunes, **M. Roosa,** A. Saastamoinen, D. P. Scriven, "Experimental Test of the Ratio Method for Nuclear-Reaction Analysis", Phys. Rev. Lett., 134, 212501, May 2025.
6. S. Ota, P. Capel, G. Christian, V. Durant, K. Hagel, E. Harris, R. C. Johnson, Z. Luo, F. M. Nunes, **M. Roosa**, A. Saastamoinen, D. P. Scriven, "First Experimental Test of the Ratio Method", Nuclear Physics A, 1060, 123120, Aug. 2025.
7. J. Bishop, G. V. Rogachev, S. Ahn, M. Barbui, S. M. Cha, E. Harris, C. Hunt, C. H. Kim, D. Kim, S. H. Kim, E. Koshchiy, Z. Luo, C. Park, C. E. Parker, E. Pollacco, B. T. Roeder, **M. Roosa**, A. Saastamoinen, D. P. Scriven, "Cluster structure of $3\alpha + p$ states in ^{13}N", Phys. Rev. C, 109, 054308, May 2024.
8. Z. Luo, M. Barbui, J. Bishop, G. Chubarian, V. Z. Goldberg, E. Harris, E. Koshchiy, C. E. Parker, **M. Roosa**, A. Saastamoinen, D. P. Scriven, G. V. Rogachev, "Radiative decay branching ratio of the Hoyle state", Phys. Rev C. 109, 025801, American Physical Society, Feb. 2024.
9. S. Ota, G. Christian, B. J. Reed, W. N. Catford, S. Dede, D. T. Doherty, G. Lotay, **M. Roosa**, A. Saastamoinen, D.P. Scriven, "BlueSTEAl: A pair of silicon arrays and a zero-degree phoswich detector for studies of scattering and reactions in inverse kinematics", Nucl. Inst. and Methods in Phys. Res. Sect. A, 1059, 168946, Feb. 2024.
10. C. Hunt, S. Ahn, J. Bishop, E. Koshchiy, E. Aboud, M. Alcorta, A. Bosh, K. Hahn, S. Han, C. E. Parker, E. C. Pollacco, B. T. Roeder, **M. Roosa**, S. Upadhyayula, A. S. Volya, and G. V. Rogachev, "Spectroscopy of ^{13}Be through isobaric analogue states in ^{13}B", Phys. Rev. C, 108, L051606, American Physical Society, Nov. 2023.
11. J. Bishop, G. V. Rogachev, S. Ahn, M. Barbui, S. M. Cha, E. Harris, C. Hunt, C. H. Kim, D. Kim, S. H. Kim, E. Koshchiy, Z. Luo, C. Park, C. E. Parker, E. C. Pollacco, B. T. Roeder, **M. Roosa**, A. Saastamoinen, and D. P. Scriven, "First Observation of the $\beta 3\alpha$p Decay of ^{13}O via β-Delayed Charged-Particle Spectroscopy", Phys. Rev. Lett. 130, 222501, June 2023.
12. L. J. Sun, J., M. Friedman, T. Budner, D. Pérez-Loureiro, E. Pollacco, C. Wrede, B. A. Brown, M. Cortesi, C. Fry, B. E. Glassman, J. Heideman, M. Janasik, A. Kruskie, A. Magilligan, **M. Roosa**, J. Stomps, J. Surbrook, & P. Tiwari, "^{25}Si β^{+}-decay spectroscopy", Phys. Rev. C, 103, 014322 American Physical Society, Jan. 2021.
13. M. Friedman, T. Budner, D. Pérez-Loureiro, E. Pollacco, C. Wrede, J. José, B. A. Brown, M. Cortesi, C. Fry, B. E. Glassman, J. Heideman, M. Janasik, **M. Roosa**, J. Stomps, J. Surbrook, & P. Tiwari, "Low-energy ^{23}Al β-delayed proton decay and ^{22}Na destruction in novae", P. Phys. Rev. C, 101, 052802, American Physical Society, May 2020.
14. M. Friedman, D. Pérez-Loureiro, T. Budner, E. Pollacco, C. Wrede, M. Cortesi, C. Fry, B. E. Glassman, M. Harris, J. Heideman, M. Janasik, B. Roeder, **M. Roosa**, A. Saastamoinen, J. Stomps, J. Surbrook, P. Tiwari, & J. Yurkon, "GADGET: a Gaseous Detector with Germanium Tagging", Nucl. Inst. and Methods in Phys. Res. Sect. A 940, 93-102, Oct. 2019.

SELECTED TALKS & POSTERS

" Strong-field QED via precision spectroscopy of antiprotonic atoms." May 2026
International Conference on Precision Physics of Simple Atomic Systems,
Marietta Blau Institute of the AAS; Vienna, Austria
Contributed Talk.

"TESs for $\bar{P}$-Atom X-ray Spectroscopy in the 100 keV regime." Oct. 2025
5^{th} International Conference on Nuclear Photonics,
TU Darmstadt; Darmstadt, Germany
Invited Talk.

"BSQED Benchmarks with $\bar{P}$-Atoms X-ray Spec. at CERN" Sept. 2025
22^{nd} Topical Workshop of the Stored Particles Atomic physics Research Collaboration,
GSI and the ExtreMe Matter Institute; Ioannina, Greece
Invited Talk.

"Transition Edge Sensors for Antiprotonic Atom X-ray Spectroscopy" Feb. 2025
Detection Systems and Techniques for fundamental and applied physics,
University of Messina & INFN; Catania, Italy.
Contributed Talk.

"Spectroscopy of ^{12}Be using TexAT-TPC" Nov. 2023
Fall Meeting of the Division of Nuclear Physics,
American Physical Society, Waikoloa Village, HI USA.
Contributed Talk.

"Spectroscopy of ^{12}Be using TexAT TPC" June 2023
Advances in Radioactive Isotope Science; Avignon, France
Poster Presentation.

"Spectroscopy of Beryllium isotopes using TexAT-TPC" Oct. 2022
Fall Meeting of the Division of Nuclear Physics,
American Physical Society; New Orleans, LA USA.
Contributed Talk.

"Spectroscopy of Beryllium isotopes using TexAT-TPC" Sept. 2022
CENTAUR Scientific Advisory Committee Meeting,
Lawrence Livermore National Lab; Livermore, CA USA.
Poster Presentation.

"Spectroscopy of ^{11,12}Be using TexAT TPC" May 2022
JINA-CEE Frontiers in Nuclear Astrophysics,
University of Notre Dame; South Bend, IN USA.
Poster Presentation.

"Shell structure and evolution through spectroscopy of ^{11}Be" Sept. 2019
CENTAUR Scientific Advisory Committee Meeting,
Los Alamos National Labl; Los Alamos NM USA
Poster Presentation.

"Simulating the Response of a New β-Delayed Proton Detector" Apr. 2018
20^{th} Annual University Undergraduate Research and Arts Forum,
MSU; East Lansing, MI USA.
Contributed Talk.

"Simulating The Response of a Gas-Filled β-Delayed Proton Detector" Oct. 2017
Fall Meeting of the Division of Nuclear Physics,
American Physical Society; Pittsburgh, PA USA.
Poster Presentation.